

Implications for prior work. Notably, the above theorems imply that three previously proposed methods in the literature, Spectral Attribute Removal (SAL) [36], Mean Projection [17], and Fair PCA [20], are guaranteed to achieve linear guardedness given suitable hyperparameters. See Appendix D for further discussion.

4.1 Derivation of LEACE

Theorem 4.1 is a very weak condition, which is far from identifying unique values for $\mathbf{P}$ and $\mathbf{b}$. In most applications, however, we’d like to make a “small” edit to $\mathbf{X}$ so that useful information contained in $\mathbf{X}$ is maximally preserved. We operationalize the notion of a small edit in terms of the mean squared norm $\mathbb{E}\|\mathbf{r}(\mathbf{X}) - \mathbf{X}\|_{\mathbf{M}}^2$ defined by some positive-definite inner product $\mathbf{M}$,⁴ which can be thought of as a local quadratic approximation to *any* measure of divergence between $\mathbf{X}$ and $\mathbf{r}(\mathbf{X})$ (such as Kullback–Leibler divergence, for example). While we are primarily interested in the Euclidean ($\mathbf{M} = \mathbf{I}$) and Mahalanobis ($\mathbf{M} = \Sigma_{\mathbf{X}\mathbf{X}}^+$) norms, it will turn out that there is a *single* erasure function that minimizes *all* such norms simultaneously. We will see in Section 6 that ensuring edits are small in this sense provides substantial benefit to downstream task performance as compared to other methods which also guard the labels $\mathbf{Z}$.

Below, we derive the optimal eraser under the assumption that $\mathbf{X}$ and $\mathbf{Z}$ are centered.

Theorem 4.2. *Let $\mathbf{X}$ and $\mathbf{Z}$ be centered random vectors taking values in $\mathbb{R}^d$ and $\mathbb{R}^k$ respectively, each of finite second moment. Let $\mathbf{M} \in \mathbb{R}^{d \times d}$ be a p.s.d. matrix defining a (possibly degenerate) inner product on $\mathbb{R}^d$: $\langle \mathbf{x}, \mathbf{y} \rangle_{\mathbf{M}} = \mathbf{x}^T \mathbf{M} \mathbf{y}$. Let $\Sigma_{\mathbf{X}\mathbf{X}} \in \mathbb{R}^{d \times d}$ be $\mathbf{X}$ ’s covariance matrix, and $\Sigma_{\mathbf{X}\mathbf{Z}} \in \mathbb{R}^{d \times k}$ be the cross-covariance matrix of $\mathbf{X}$ and $\mathbf{Z}$. Let $\mathbf{A}^+$ denote the Moore–Penrose pseudoinverse of a matrix $\mathbf{A}$, and let $\mathbf{A}^{1/2}$ be the p.s.d. square root of a p.s.d. matrix $\mathbf{A}$. Then the objective*

$$\operatorname{argmin}_{\mathbf{P} \in \mathbb{R}^{d \times d}} \mathbb{E} \left[\|\mathbf{P}\mathbf{X} - \mathbf{X}\|_{\mathbf{M}}^2 \right] \quad \text{subject to } \operatorname{Cov}(\mathbf{P}\mathbf{X}, \mathbf{Z}) = \mathbf{0}$$

has the following solution:

$$\mathbf{P}^* = \mathbf{I} - \mathbf{W}^+ \mathbf{P}_{\mathbf{W}\Sigma_{\mathbf{X}\mathbf{Z}}} \mathbf{W},$$

where $\mathbf{W}$ is the whitening transformation $(\Sigma_{\mathbf{X}\mathbf{X}}^{1/2})^+$ and $\mathbf{P}_{\mathbf{W}\Sigma_{\mathbf{X}\mathbf{Z}}} = (\mathbf{W}\Sigma_{\mathbf{X}\mathbf{Z}})(\mathbf{W}\Sigma_{\mathbf{X}\mathbf{Z}})^+$ is the orthogonal projection matrix onto $\operatorname{colsp}(\mathbf{W}\Sigma_{\mathbf{X}\mathbf{Z}})$.

Proof. See Appendices E.1 and E.2 for two independent proofs of Theorem 4.2. □

The above theorem assumes that the random vectors $\mathbf{X}$ and $\mathbf{Z}$ are centered, and does not include a bias term. Below we extend our results to the uncentered case, and derive the optimal bias $\mathbf{b}^*$.

Theorem 4.3. *Let $\mathbf{X}$ and $\mathbf{Z}$ be random vectors taking values in $\mathbb{R}^d$ and $\mathbb{R}^k$ respectively, each of finite second moment. Define $\mathbf{M}$ and $\mathbf{P}^*$ as in Theorem 4.2 and $\mathbf{b}^* = \mathbb{E}[\mathbf{X}] - \mathbf{P}^* \mathbb{E}[\mathbf{X}]$. Then $(\mathbf{P}^*, \mathbf{b}^*)$ minimizes $\mathbb{E}\|\mathbf{P}\mathbf{X} + \mathbf{b} - \mathbf{X}\|_{\mathbf{M}}^2$, subject to $\operatorname{Cov}(\mathbf{P}\mathbf{X} + \mathbf{b}, \mathbf{Z}) = \mathbf{0}$.*

Proof. Let $\mathbf{P} \in \mathbb{R}^{d \times d}$ and define $\tilde{\mathbf{X}} = \mathbf{X} - \mathbb{E}[\mathbf{X}]$ and $\mathbf{c} = \mathbf{P}\mathbb{E}[\mathbf{X}] + \mathbf{b} - \mathbb{E}[\mathbf{X}]$. Then,

$$\begin{aligned} \mathbb{E}\|\mathbf{P}\mathbf{X} + \mathbf{b} - \mathbf{X}\|_{\mathbf{M}}^2 &= \mathbb{E}\|(\mathbf{P}\tilde{\mathbf{X}} - \tilde{\mathbf{X}}) + \mathbf{c}\|_{\mathbf{M}}^2 \\ &= \mathbb{E}\|\mathbf{P}\tilde{\mathbf{X}} - \tilde{\mathbf{X}}\|_{\mathbf{M}}^2 + 2\mathbb{E}[\mathbf{P}\tilde{\mathbf{X}} - \tilde{\mathbf{X}}]^T \mathbf{M} \mathbf{c} + \mathbf{c}^T \mathbf{M} \mathbf{c} \\ &= \mathbb{E}\|\mathbf{P}\tilde{\mathbf{X}} - \tilde{\mathbf{X}}\|_{\mathbf{M}}^2 + \mathbf{c}^T \mathbf{M} \mathbf{c}, \end{aligned}$$

where we have eliminated the middle term because $\mathbf{P}$ is linear and $\mathbb{E}[\tilde{\mathbf{X}}] = \mathbf{0}$. Since $\mathbf{M}$ is p.s.d., our objective is minimized for $\mathbf{c} = \mathbf{0}$, i.e. $\mathbf{b} = \mathbb{E}[\mathbf{X}] - \mathbf{P}\mathbb{E}[\mathbf{X}]$. The problem thus reduces to choosing $\mathbf{P}$ so as to minimize $\mathbb{E}\|\mathbf{P}\tilde{\mathbf{X}} - \tilde{\mathbf{X}}\|_{\mathbf{M}}^2$ subject to $\operatorname{Cov}(\mathbf{P}\mathbf{X} + \mathbf{b}, \mathbf{Z}) = \operatorname{Cov}(\mathbf{P}\tilde{\mathbf{X}}, \mathbf{Z}) = \mathbf{0}$, which Theorem 4.2 shows occurs when $\mathbf{P} = \mathbf{P}^*$. □

⁴Our proofs also include degenerate “inner products” where $\mathbf{M}$ is singular, and the associated seminorms.

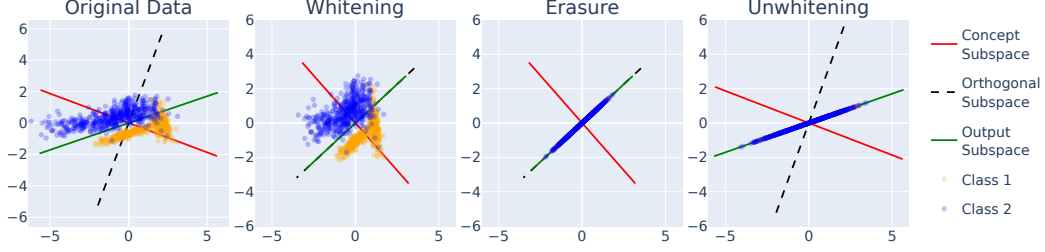


Figure 1: LEACE projection in 3 steps. First the data is whitened, ensuring equal variance in all directions. It is then orthogonally projected onto $\text{colsp}(\mathbf{W}\Sigma_{XZ})^\perp$, guaranteeing linear guardedness. Finally, we unwhiten the data so that its covariance structure mimics the original.

Putting together Theorems 4.2 and 4.3 and rearranging, we arrive at the LEACE formula:

$$r_{\text{LEACE}}(\mathbf{x}) = \mathbf{x} - \mathbf{W}^+ \mathbf{P}_{\mathbf{W}\Sigma_{XZ}} \mathbf{W}(\mathbf{x} - \mathbb{E}[\mathbf{X}]) \quad (1)$$

Intuitively, LEACE de-means and whitens $\mathbf{x}$, projects onto the subspace responsible for correlations between X and Z , then unwhitens the result. Finally, it subtracts this value from $\mathbf{x}$, thereby surgically removing the linearly available information about Z .

4.2 Oblique Projections are Least-Squares Optimal

Prior work on linear concept erasure has assumed that erasure functions should be orthogonal projections [29, 32, 36], appealing to the well-known fact that an orthogonal projection of a point $\mathbf{x}$ onto a subspace U yields the nearest point in U to $\mathbf{x}$. But even in the case where X is centered, r_{LEACE} is *not* an orthogonal projection in general. Orthogonal projection matrices are symmetric, and $\mathbf{I} - \mathbf{W}^+ \mathbf{P}_{\mathbf{W}\Sigma_{XZ}} \mathbf{W}$ is only symmetric in the special case where $\mathbf{P}_{\mathbf{W}\Sigma_{XZ}}$ and $\mathbf{W}$ commute. It is an *oblique* projection however, since applying $\mathbf{P}^*$ twice yields the same result as applying it once: $(\mathbf{P}^*)^2 = \mathbf{I} - 2\mathbf{W}\mathbf{P}_{\mathbf{W}\Sigma_{XZ}}\mathbf{W}^+ + \mathbf{W}^+ \mathbf{P}_{\mathbf{W}\Sigma_{XZ}} \mathbf{W}\mathbf{W}^+ \mathbf{P}_{\mathbf{W}\Sigma_{XZ}} \mathbf{W} = \mathbf{P}^*$.

Orthogonal projections are generally not least-squares optimal for concept erasure because the necessary and sufficient condition for linear guardedness, $\mathbf{P}\Sigma_{XZ} = \mathbf{0}$, is a constraint on the *nullspace* of $\mathbf{P}$, and not on its range. We may freely choose the range of the projection to minimize the mean squared distance, as long as we zero out $\text{colsp}(\Sigma_{XZ})$. In Figure 1, an orthogonal projection would map all points onto the dashed line, thereby preserving less of the variance of the original data than LEACE does (green line). See Appendix F for a concrete example.

4.3 Extension to Continuous Z

While not a focus of this work, it’s worth noting that LEACE can also be applied to the setting where Z takes arbitrary values in $\mathbb{R}^k$, as long as we restrict ourselves to the ordinary least squares regression loss $\mathcal{L}(\eta, \mathbf{z}) = \|\eta - \mathbf{z}\|_2^2$. In particular, the proofs of equivalence between conditions 1 and 2 given in Appendix B make no categorical assumption on Z , and the equivalence between the optimality of a zero weight matrix (condition 2) and zero cross-covariance (condition 4) is well known in the OLS setting. We can then apply Theorems 4.2 and 4.3, which also make no categorical assumption, to derive the same optimal affine eraser as in the categorical case.

5 Evaluation

5.1 Intrinsic Evaluation

Following Ravfogel et al. [31] we evaluate the ability of our method to remove gender information from the last hidden layer of a frozen BERT model. We use the biographies dataset of De-Arteaga et al. [6], composed of short biographies annotated by both binary gender and profession. We embed each biography with the [CLS] embedding in the last layer of BERT, enforce the same-conditional-mean constraint to remove gender information from the [CLS], and then evaluate the performance of the model, after the intervention, on the main task of profession prediction. We

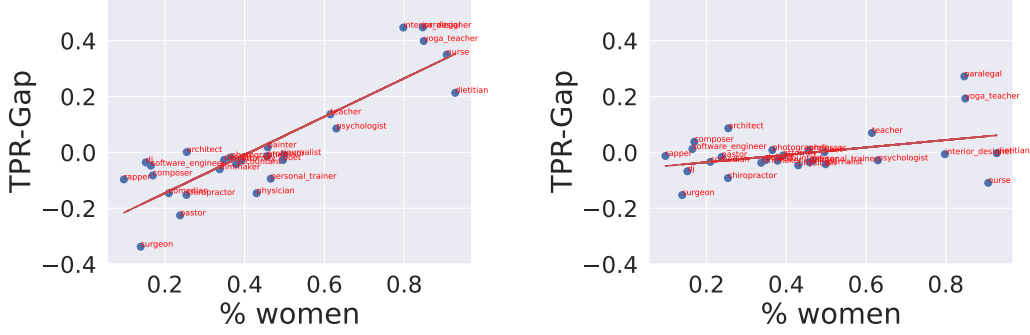


Figure 3: The correlation between $GAP_{female,y}^{TPR}$ and the relative proportion of women in profession y , for BERT embedding, before (left; $R=0.867$) and after (right; $R=0.392$) the projection.

compare our intervention with RLACE [31], which uses gradient-based optimization to solve a linear concept-erasure adversarial game.

Concept erasure results. First, we evaluate the ability of logistic regression classifiers to recover the removed information. The results, presented in Fig. 2, show that our method is the only to achieve random accuracy (perfect erasure) with a small edit, although RLACE (but not INLP) comes close. At the same time, our method is around 2 orders of magnitude faster, and does not require gradient-based optimization.

5.2 Downstream Fairness

How does our intervention affect the behavior of the model on the main classification task of profession prediction? We fit a logistic regression profession-prediction classifier over the projected [CLS] embeddings.

To measure the bias in a classifier, we follow De-Arteaga et al. [6] and use the TPR-GAP measure, which quantifies the bias in a classifier by considering the difference (GAP) in the true positive rate (TPR) between individuals with different protected attributes (e.g. race or gender). We use the notation $GAP_{z,y}^{TPR}$ to denote the TPR-gap in some main-class label y (e.g. “nurse” prediction) for some protected group z (e.g. “female”), we also consider $GAP_z^{TPR,RMS}$, the RMS of the TPR-gap across all professions for a protected group z :

$$GAP_z^{TPR,RMS} = \sqrt{\frac{1}{|C|} \sum_{y \in C} (GAP_{z,y}^{TPR})^2}$$

To calculate the relation between the bias the model exhibits and the bias in the data, we also calculate $\sigma_{(GAP^{TPR}, \%Women)}$, the correlation between the TPR gap in a given profession and the percentage of women in that profession.

Results. The main-task classifier achieves profession-prediction accuracy of 77.3% on the projected embeddings (compared with 79.3% over the original embeddings), indicating that the intervention minimally affects the ability to predict the profession of a person from the embedding of their biography. At the same time, the TPR gap drops significantly from 0.198 to 0.084, indicating a sharp drop in the biased behavior of the profession classifier. Indeed, inspecting the correlation $\sigma_{(GAP^{TPR}, \%Women)}$ between the gap (per profession) and the embedding of women in this profession,

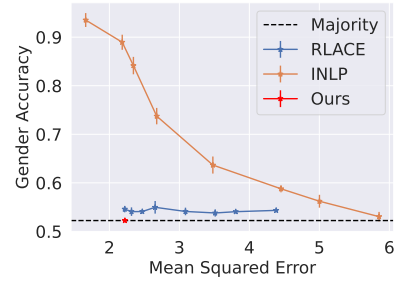


Figure 2: Gender prediction accuracy after bias-removal projection versus the mean squared distance from the original embedding for INLP, RLACE, and LEACE on BERT embeddings.

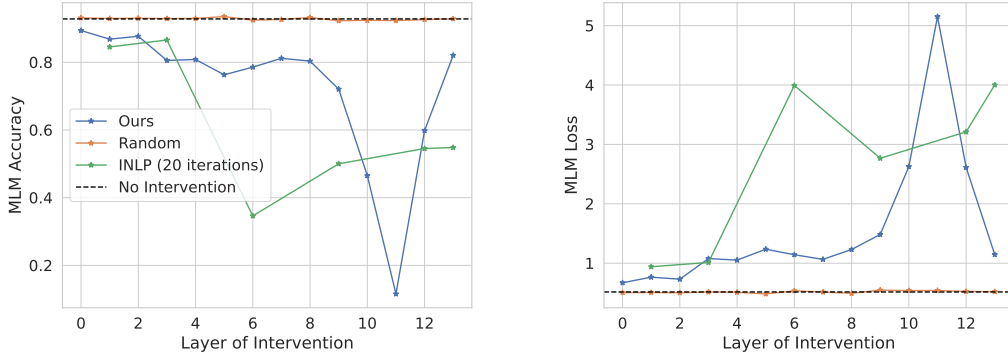


Figure 4: Amnesic probing results on bert-base-uncased.

we see that this correlation plummets from 0.867 to 0.392 after erasure. Re-fitting the main-task logistic regression classifier over the projected embeddings yields a slightly higher main-task accuracy of 78.1%, at the price of significantly increasing the TPR gap to 0.158.⁵

5.3 Revisiting Amnesic Probing

Elazar et al. [10] have introduced the idea of *amnesic probing* as a causal intervention that aims to test the importance of a given concept (e.g. part-of-speech tag) to some main task (e.g. language modeling). They applied Iterative Nullspace Projection (INLP) to remove different concepts from the activations of the model, and assessed the degree to which its behavior changed when performing masked language modeling. Since INLP often requires dozens of iterations to completely erase the concept, its usage in this context raises concerns of collateral damage due to magnitude of the intervention and the non-exhaustive nature of INLP removal. Here, we replicate their experiments on the bert-base-uncased model with our interventions.

Experimental setup. We use part-of-speech (POS) tags as our concept of interest. We collect sentences and their coarse POS tags (“Noun”, “Verb” etc.; 18 in total) from the English Universal Dependencies dataset [27]. We tokenize the sentences with the BERT tokenizer and map each word-piece to the POS tag of the word to which it belongs. We collect the unmasked BERT embeddings for each layer, intervene to linearly erase the POS concept from that layer, and continue the forward pass until the last layer, from which we compute the distribution of the MLM over the vocabulary. Note that in each experiment we intervene on a single layer. We quantify the decrease in accuracy following the intervention, as well as the increase in the loss. We compare with a baseline intervention of a random orthogonal projection whose null space has the same rank as the label space (18). For INLP, we perform 20 iterations. This is needed because INLP does not effectively remove the concept; even after 20 iterations, classification accuracy is above majority accuracy. As a result, INLP reduces the rank of the embedding by 360. By contrast, our method decreases the rank just by 17.

Results. The results are shown in Fig. 4b. Our intervention only mildly changes BERT LM accuracy and loss until layer 8, with the highest drop recorded in layer 11. INLP, in contrast, shows maximum effect at layer 6. Since it removes hundreds of dimensions, it is difficult to attribute this effect to the erasure of the concept. These results suggest that the *causal* effect of the POS concept on the language model is concentrated in layer 11. Interestingly, this stands in contrast with POS linear probing results, which are optimal at earlier layers [38]. As Elazar et al. [10] have noted, probing does not generally correlate with intervention-based analysis techniques.

⁵The softmax probabilities of a multiclass logistic regression classifier can leak the removed information if another classifier is stacked on top of it [33], though this setup is not linear.